PROJECT PROPOSAL PRESENTATION · IT4010 RESEARCH PROJECT · 2026

AI Based Emergency Call and SMS Triage System for Disaster Response in Sri Lankan Low Resource Local Languages

R26-DS-011



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Overall Project Description

Introduction · Research problem and question · Objectives · Overall system diagram

Introduction to the Overall Project

Sri Lanka faces recurring floods, landslides and fires. During a disaster, people report emergencies by phone call and SMS, in Sinhala, Tamil, English or a mix of them (Singlish, Tanglish).

Reports are often panicked, incomplete and noisy, and many people report the same incident in different words.

Our project builds one AI pipeline that supports the Disaster Management Centre operator from the first call or SMS to a verified incident.

Today, an operator must do all three by hand:

1. **Understand**
   multilingual, sometimes panicked speech and broken SMS
2. **Judge urgency**
   decide how urgent each report really is
3. **Detect duplicates**
   notice when several reports describe one incident

None of these judgements is automated for Sri Lankan languages today.

Research Problem and Research Question

Language
Speech and text models are built for English; Sinhala, Tamil and code-mixed input are barely supported

Incomplete reports
Callers and SMS senders leave out location, people count or danger, and systems do not ask

Urgency
Urgency is judged from words only, or by hand; vocal stress is ignored

Duplicates
Calls and SMS are handled separately, so one incident can be dispatched twice while another is missed

RESEARCH QUESTION

How can an AI-based system understand multilingual emergency calls and SMS, clarify missing information, prioritise reports by urgency and fuse duplicates into one verified incident for faster disaster response in Sri Lanka?

Objectives

MAIN OBJECTIVE **Design, build and evaluate an integrated AI triage system that turns multilingual emergency calls and SMS into prioritised, de-duplicated and verified incident records.**

C1 **Amanda Edirisinghe**

Ask adaptive follow-up questions in the caller's language until the call report is dispatch-ready

C2 **Shilma Thanseel**

Classify call urgency (High / Medium / Low) from the caller's voice and words

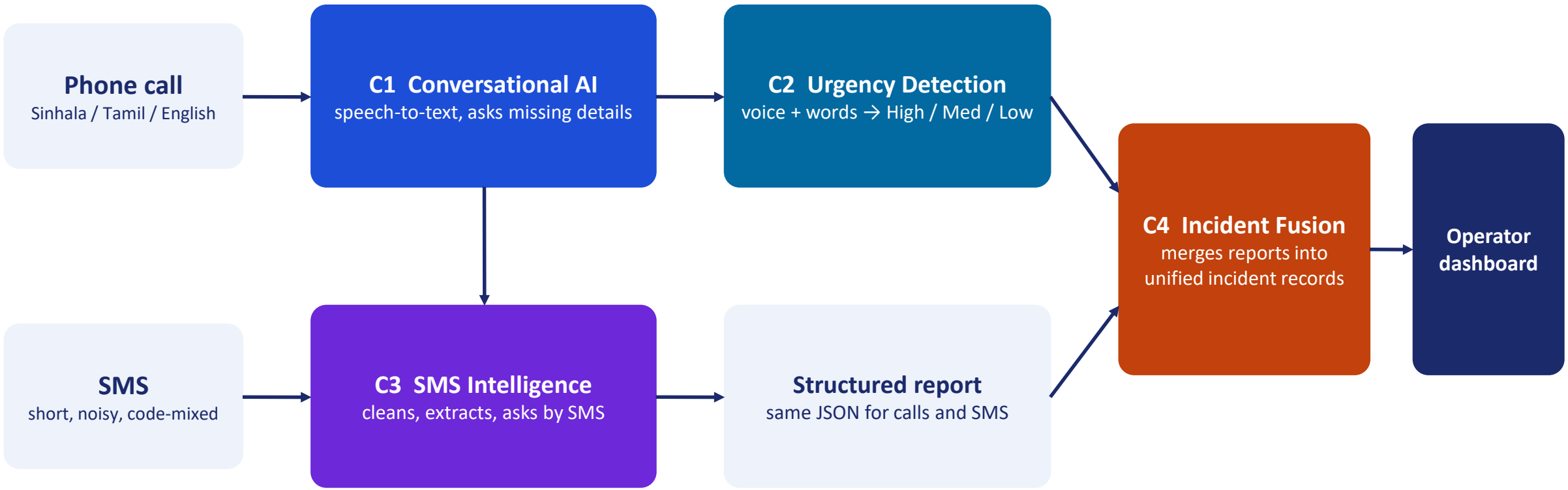
C3 **Gawrangana Karunarathna**

Turn short, noisy, code-mixed SMS into structured, prioritised emergency intelligence

C4 **Amandhi Navoda Siriwardena**

Fuse call and SMS reports into unified incident records on a real-time operator dashboard

Overall System Diagram



One pipeline: multilingual calls and SMS in, one verified, prioritised incident out for the dispatcher.

C1

INDIVIDUAL COMPONENT

Multilingual Emergency Conversational AI System for Rapid Clarification



Amanda Edirisinghe

IT22129758 · Data Science

Detect missing information

Ask the next question

Sinhala · Tamil · English · code-mixed

Research Question, Specific and Sub Objectives

RESEARCH QUESTION **How can a multilingual AI agent detect missing critical information in Sinhala, Tamil, English and code-mixed emergency calls, and ask the right follow-up question in real time?**

Specific objective

Build a multilingual AI agent that quickly identifies and clarifies missing critical information from emergency callers, producing an accurate and complete structured summary within 30 seconds per call.

Sub objectives

1. Detect incomplete information (location, head count, immediate danger) in real time
2. Generate dynamic questions that respond to panic, confusion and language switching
3. Reach at least 90% completeness of the emergency information gathered
4. Stay robust to call drops and unexpected replies; output a consistent JSON summary
5. Evaluate time-to-complete, completeness rate, satisfaction and adaptability

Methodology

1

Missing-information detection

Check the structured call data for missing or uncertain fields

2

Dynamic question generation

Short, context-specific questions by language, emotion and urgency

3

Language & code-mix handling

Transformer models (mBERT, XLM-R) for Sinhala, Tamil, English

4

Emotion & behaviour adaptation

Rule-based prosody and dialogue management for panicked callers

5

Structured summary

Clean JSON report for the operator

Research approach

Design Science Research: iterative design and testing of the AI agent

Data

Simulated emergency calls from consenting volunteers, plus TTS, noise and code-mix augmentation

Tools

Python, Hugging Face Transformers, Whisper, Flask / FastAPI, Google Colab

Completion of the Project

Completed for the proposal

- ✓ Literature review
- ✓ Research gap and research question
- ✓ Main and sub objectives
- ✓ Methodology and tools selected
- ✓ Dataset and evaluation plan
- ✓ Work breakdown and Gantt chart

Plan to completion

M1–2	Planning: literature review, requirements, architecture, dataset plan
M3–5	**Clarification agent: question generation, multilingual support, call-drop handling, summaries**
M6–7	Incident fusion and dashboard (C4)
M8	System integration of all four components
M9	System testing and evaluation
M10	Dashboard improvement, documentation, final report

Coloured rows: this member's main development period

C2

INDIVIDUAL COMPONENT

Multimodal Urgency Detection Using Prosodic and Conversational Cues



Shilma Thanseel

IT22644480 · Data Science

How the caller speaks

What the caller says

High · Medium · Low urgency

Research Question, Specific and Sub Objectives

RESEARCH QUESTION **Can combining how a caller speaks (prosody) with what they say (conversational cues) classify Sinhala and Tamil emergency calls by urgency more reliably than either alone?**

Specific objective

Develop a multimodal urgency detection module that classifies emergency calls into High, Medium and Low urgency from prosodic features and conversational cues in Sinhala and Tamil.

Sub objectives

1. Extract prosodic features (pitch, intensity, jitter, shimmer, speech rate) with openSMILE / GeMAPS
2. Detect urgency keywords and patterns in Whisper transcripts
3. Design a hybrid fusion model that gives one urgency score in real time
4. Build a simulated multilingual call dataset; add real calls if authorised
5. Measure accuracy, precision, recall, F1 and false negatives on High-urgency calls
6. Integrate the module with the rest of the triage workflow

Methodology

1

Audio preprocessing

Noise reduction, normalisation, framing

2

Speech-to-text

Whisper ASR for Sinhala and Tamil speech

3

Prosodic features

openSMILE + GeMAPS: pitch, intensity, jitter, shimmer, rate

4

Conversational cues

Rule-based urgency keywords and severity phrases

5

Fusion & classification

Combined feature vector, ML classifier: High / Medium / Low

Research approach

Applied, experimental research with interpretable, low-resource methods

Data

Simulated Sinhala / Tamil emergency calls, real calls only through authorised channels

Tools

Python, Whisper, openSMILE, librosa, scikit-learn

Completion of the Project

Completed for the proposal

- ✓ Literature review
- ✓ Research gap and research question
- ✓ Main and sub objectives
- ✓ Methodology and tools selected
- ✓ Dataset and evaluation plan
- ✓ Work breakdown and Gantt chart

Plan to completion

M1–2	Planning: literature review, requirements, architecture, dataset plan
M3–5	**Urgency detection: preprocessing, transcription, prosody, cue detection, fusion**
M6–7	Incident fusion and dashboard (C4)
M8	System integration of all four components
M9	System testing and evaluation
M10	Dashboard improvement, documentation, final report

Coloured rows: this member's main development period

C3

INDIVIDUAL COMPONENT

Low-Bandwidth SMS Emergency Intelligence for Disaster Response



Gawrangana Karunarathna

IT22263476 · Data Science

Short, noisy SMS

Structured intelligence

Works on low bandwidth

Research Question, Specific and Sub Objectives

RESEARCH QUESTION **How can short, noisy and code-mixed emergency SMS be turned into structured, prioritised disaster intelligence when connectivity is poor?**

Specific objective

Design, develop and evaluate a low-bandwidth text intelligence module that converts short, noisy, multilingual SMS into structured emergency intelligence with urgency prioritisation.

Sub objectives

1. Detect Sinhala, Tamil, English and code-mixed text in 3 to 5 word messages
2. Normalise abbreviations, emojis, repeated letters and Romanised text
3. Extract incident type, location clues, people affected and resources needed
4. Classify urgency (Critical, High, Medium, Low) with a hybrid rule + ML model
5. Ask a few critical follow-up questions by SMS when information is missing
6. Output structured JSON for dashboards and the fusion component

Methodology

1

Language & script detection

Character n-grams, script identification, code-mix handling

2

Text normalisation

Abbreviations, emojis, repeated letters, Romanised words

3

Incident & entity extraction

NER plus rule-based patterns

4

Urgency classification

Hybrid rules + machine learning, four levels

5

SMS follow-up agent

Asks only for missing critical fields, then outputs JSON

Research approach

Design Science Research with experimental evaluation

Data

Simulated and real emergency SMS covering six language styles

Tools

Python, NLP libraries, scikit-learn, Flask, Twilio SMS

Completion of the Project

Completed for the proposal

- ✓ Literature review
- ✓ Research gap and research question
- ✓ Main and sub objectives
- ✓ Methodology and tools selected
- ✓ Dataset and evaluation plan
- ✓ Work breakdown and Gantt chart

Plan to completion

M1–2	Planning: literature review, requirements, architecture, dataset plan
M3–5	**SMS intelligence: language detection, normalisation, extraction, urgency, follow-up agent**
M6–7	Incident fusion and dashboard (C4)
M8	System integration of all four components
M9	System testing and evaluation
M10	Dashboard improvement, documentation, final report

Coloured rows: this member's main development period

C4

INDIVIDUAL COMPONENT



Multi-Source Urgency-Based Incident Fusion, Unified Incident Records and Real-Time Operator Dashboard

Amandhi Navoda Siriwardena

IT22504418 · Data Science

Calls + SMS → one incident

Unified Incident Records

Real-time operator dashboard

Research Question, Specific and Sub Objectives

RESEARCH QUESTION **How can multilingual call and SMS reports of the same disaster incident be fused into one unified incident record without merging separate emergencies?**

Specific objective

Design and evaluate a multilingual multi-source incident fusion system that de-duplicates Sinhala, Tamil and English reports into unified incident records, with precision above 90%, a critical false-merge rate below 2% and a real-time dashboard.

Sub objectives

1. Build a Sri Lankan location gazetteer of 5,000+ name variants in all scripts
2. Score report pairs on meaning (LaBSE), place (Haversine) and time
3. Tune urgency-sensitive clustering so critical incidents are rarely wrongly merged
4. Detect and resolve conflicting fields inside merged reports
5. Deliver a real-time map dashboard with operator confirm / split / merge actions
6. Evaluate on 20 multi-caller scenarios and an operator drill (SUS feedback)

Methodology

1
Normalise reports

One canonical schema for call and SMS outputs

2
Resolve locations

Multilingual gazetteer to GPS coordinates

3
Similarity scoring

LaBSE semantic + geographic + temporal

4
Urgency-sensitive clustering

Merge or separate, stricter for critical incidents

5
Unified record & dashboard

Conflict resolution, timeline, operator actions

Research approach

Design Science Research: relevance, design and rigour cycles

Data

Multilingual multi-caller disaster scenarios with ground-truth incident labels

Tools

Python, LaBSE, scikit-learn, Flask / WebSocket, Leaflet / OpenStreetMap

Completion of the Project

Completed for the proposal

- ✓ Literature review
- ✓ Research gap and research question
- ✓ Main and sub objectives
- ✓ Methodology and tools selected
- ✓ Dataset and evaluation plan
- ✓ Work breakdown and Gantt chart

Plan to completion

M1–2	Planning: literature review, requirements, architecture, dataset plan
M3–5	Components C1, C2 and C3
M6–7	**Incident fusion: clustering, duplicate detection, conflict resolution, unified records, dashboard**
M8	System integration of all four components
M9	System testing and evaluation
M10	Dashboard improvement, documentation, final report

Coloured rows: this member's main development period

References

[1] A. Radford et al., "Robust speech recognition via large-scale weak supervision," in Proc. ICML, 2023.

[2] F. Eyben, M. Wöllmer and B. Schuller, "openSMILE: the Munich versatile and fast open-source audio feature extractor," in Proc. ACM Multimedia, 2010.

[3] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale," in Proc. ACL, 2020.

[4] F. Feng, Y. Yang, D. Cer, N. Arivazhagan and W. Wang, "Language-agnostic BERT sentence embedding," in Proc. ACL, 2022.

[5] M. Imran, C. Castillo, F. Diaz and S. Vieweg, "Processing social media messages in mass emergency: A survey," ACM Comput. Surv., vol. 47, no. 4, pp. 1–38, 2015.

[6] R. Lamsal, M. A. Hasnat, A. Nayak and M. Imran, "CrisisTransformers: Pre-trained language models and sentence encoders for crisis-related social media texts," Knowledge-Based Syst., vol. 296, 2024.

[7] A. Tahernejad, A. Sahebi, A. S. Sahl Abadi and M. Safari, "Application of artificial intelligence in triage in emergencies and disasters: a systematic review," BMC Public Health, vol. 24, art. 3203, 2024.

[8] A. Mukhopadhyay et al., "A review of incident prediction, resource allocation, and dispatch models for emergency response management," Accid. Anal. Prev., vol. 165, 2022.

Thank You

Questions? · R26-DS-011 · sahaya.researchteam@gmail.com